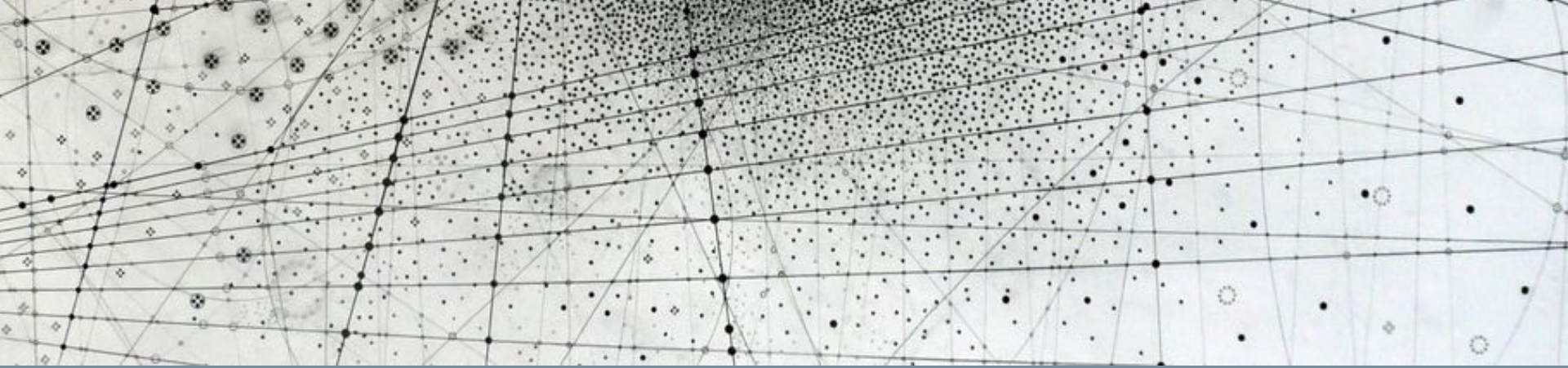




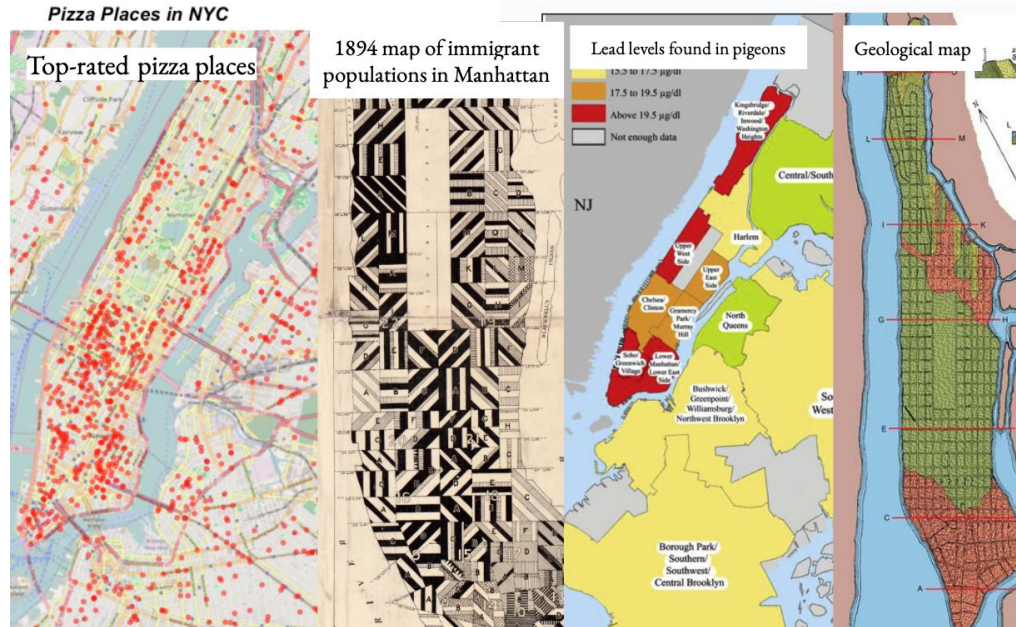
Research Methods in Machine Learning & Physics

Quick recap

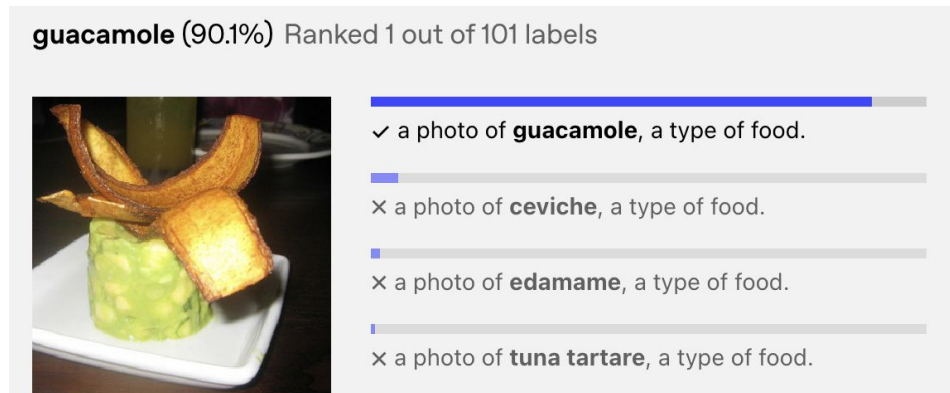
Multimodal models

A **modality** is a specific way of perceiving the world, sort of like our five senses.

In ML, "multimodality" refers to **heterogeneous & interconnected** data sources.
Common modalities include text, audio, video, etc.



AI industry has multimodal models, but the modalities are relatively few and well-defined:

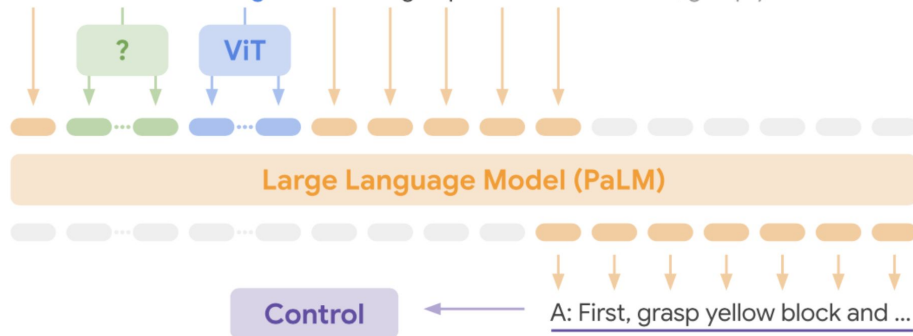


Text & images

(OpenAI's CLIP, 2021)

PaLM-E: An Embodied Multimodal Language Model

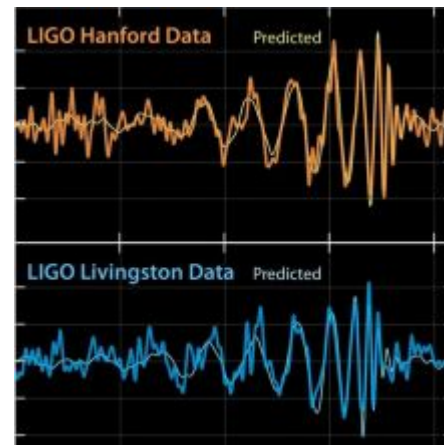
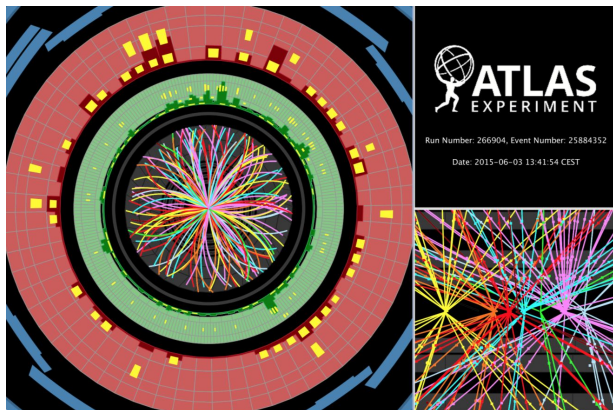
Given **<emb>** ... **** Q: How to grasp blue block? A: First, grasp yellow block



Text & images &
robotic instructions

(Driess et. al., [PaLM-E](#), 2023)

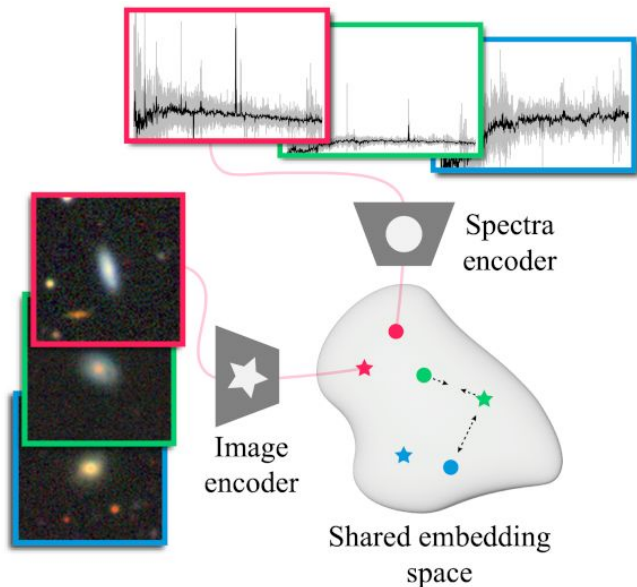
Physics data, on the other hand, is highly diverse and multimodal.



Common embedding?

Multimodal models see promising performance in industry and in science.

Significant evidence suggests that **including multiple views/modalities** in a single model can improve performance on downstream tasks relative to unimodal models:



	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

(CLIP, 2021)

Simulation-based Inference (SBI)

Data Science Institute Director Kyle Cranmer awarded inaugural prize for AI in science

Posted on [November 10, 2025](#)

Kyle Cranmer, professor of physics and director of the Data Science Institute at UW–Madison, has been honored with the inaugural [Pritzker Prize in AI for Science](#) for his role in establishing a powerful new approach to data analysis that leverages advances in machine learning and artificial intelligence.

Cranmer received this award on Monday at the Conference on AI+Science, hosted by Caltech and the University of Chicago, in Pasadena. The prize recognizes his contributions to and advocacy for [simulation-based inference](#), which combines AI and computer simulations to enable statistical analysis of data in situations where it was not previously possible.

“With advances in machine learning and AI, simulation-based inference is ushering in a new era of statistical inference for many different areas of science, and I find it extremely compelling,” says Cranmer.



Kyle Cranmer received the Pritzker Prize for AI in Science in recognition of his contributions to simulation-based inference.

Talk recording:

<https://www.youtube.com/watch?v=gEd6asXZP10&t=11805s>

A decorative graphic in the top-left corner consisting of a network of interconnected nodes and lines, resembling a molecular or computational structure.

The Caltech and University of Chicago
Conference on
AI+Science

Sponsored by the Marshall and Tom Phipps Foundations

A colorful abstract banner at the bottom of the slide, featuring geometric shapes and a network-like pattern in shades of blue, green, and orange.

In-class work

Apply the SBI package to extract unknown parameters from a dataset of 5 data points... the group with the closest guess wins :)

"Simulator" = 2D Gaussian, where each Gaussian has $\mu = ?$ and $\sigma = 1$.

References:

- `sbi` package documentation
- SBI tutorial by Miles Cranmer: <https://astroautomata.com/blog/simulation-based-inference/>
- SBI: A Practical Guide (<https://arxiv.org/html/2508.12939v1>)

We'll reconvene at the top of the hour to share our highlights!